



Approved by Chair:

A handwritten signature in black ink, appearing to read 'A. Davis', with a horizontal line underneath.

19 August 2025

Signature

COMP3104 - DevOps

Course Description

In this course students will learn the procedures, protocols and best practices that a DevOps engineer brings to a workplace environment to increase productivity. Analyze and evaluate various developer operation implementations available and implement hands-on work using industry standard toolkit. Create solutions to complex business problems by adhering to the core responsibilities of DevOps.

Course Outcomes

At the end of this course, the student will reliably demonstrate the ability to:

1. Demonstrate knowledge of effective DevOps techniques and be able to provide solutions using industry standard tools and techniques such as continuous integration/delivery/deployment strategies, software development lifecycle, application containerization for deployment, etc.
2. Perform hand-on lab work by applying DevOps techniques.
3. Ensure effective cross collaboration amongst IT staff by following the role and responsibilities of an engineer.
4. Analyze and utilize Version Control System workflows for low-risk software releases.
5. Take well informed architectural decisions for software products, specifically in the context of monolith vs microservice architectures.
6. Monitor code quality and deploy applications to hosting services.
7. Utilize service platforms such as Infrastructure as Code and the associated benefits of cloud computing products.
8. Identify the market space as a DevOps engineer. s.

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Required:

- Exploring DevOps: An Introduction for Beginners - 978-1-927041-22-2

Recommended Resources:

1. <https://github.com>
2. <https://travis-ci.com>
3. <https://docker.com>
4. <https://kubernetes.io>
5. <https://www.sonarqube.org>

Course Delivery Mode

The course uses various instructional methods, such as lectures, demonstrations, hands-on exercises, and take-home assignments. The delivery mode depends on whether the course is online or in person. Online lectures will be the primary mode, but there may be in-person lectures for in-person participants. Labs will be conducted virtually for the online program, while in-person program students must attend on-campus labs. For more information about the delivery mode, please refer to D2L. Any updates will be communicated through D2L in advance.

Assignment Policy

- All assignments must be submitted on the due date based on an instruction given by the professor. Late assignment will be penalized 20% per day to a maximum of seven days, the weekend included unless the student has notified the professor (via email, phone or in person) ahead of the due date that he/she has a valid reason for a late submission.

Test Policy

- Students must complete tests and the final exam on the assigned day. If unable to complete the test/exam as scheduled, students are required to notify the professor at least three days prior to the date, so alternative arrangements can be made. Failure to comply with this policy may result in a zero grade.
- There will be no makeup quiz and lab exercises. If you anticipate missing more than two quizzes or lab exercises for serious, major reasons, notify your professor beforehand.

In-Person Exam Policy

Mid-term and Final exams for the T177 program will be conducted in person. Please note the following exam schedule:

- Mid-Term Exams: Week 7 of the semester
- Final Exams: Week 15 of the semester

Students are expected to be available in person during these exam periods.

Important Note on the Use of Generative AI:

Students must review the "Generative AI Usage Guidelines" document, available on D2L, for detailed instructions on how generative AI tools may be used in this course. The course evaluation table now includes a column labelled "AI Usage Allowed," indicating whether AI use is permitted for each assessment.

Yes: AI can be used with proper referencing.

No: AI cannot be used, and any usage will be considered plagiarism and subject to academic penalties.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity

EVALUATION SYSTEM:

The passing grade for this course is: D (50%)

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:	AI Usage Allowed
Lab Exercises	Completion of lab exercises 8 out of 10	1,2,3,4,5,6,7	1,2,4,5,6,7,8,9,10,11	1-6 and 9-14	16	NO
Lecture Quizzes	10 quizzes, best 8 out of 10 will be considered	1,2,3,4,5,6,7,8	2,3,4,5,6,7,10,11	1-6 and 9-14	8	NO
Assignment	Individual assignment	1,2,3,4,5,6,7	1,2,4,5,6,7,8,9,10,11	12	10	YES
Assignment	Group assignment	1,2,3,4,5,6,7	1,2,4,5,6,7,8,9,10,11	6	16	YES
Exam	Mid-term exam	1, 2, 3,4,5,6	1,2,4,5,6,7,10,11	7	20	NO
Exam	Final exam	1,2,3,4,5,6,7,8	1,2,4,5,6,7,10,11	15	30	NO
				TOTAL	100%	

Topical Outline

Learning Schedule / Topical Outline (subject to change with notification)

Week	Topic / Task	Outcomes	Content / Activities	Resources
1	Intro, BASH & The CLI	1-2	- Why DevOps? - Future of DevOps - What is a build pipeline? - Introduction to course material - Project Assignment Overview - BASH Commands 101 - Installing Course Dependencies	Lab, Lecture, Exercise & Supplementary Material

2	Git, GitHub & more BASH	1-2	- Version Control Systems - Utilization of VCS - VCS Terminology - Role of VCS in build pipelines - CVCS vs DVCS - GitHub Overview - More BASH commands and shell script - Asymmetric crypto, SSH and Access token - CLI Text Editors - Configuring your local environment for VCS - o GitHub Desktop - o SourceTree	Lab, Lecture, Exercise & Supplementary Material
3	Version Control for DevOps	1-2, 5	- Leveraging VCS for DevOps - Application Architectures, then & now - Monolith vs Microservice architecture - Vertical & horizontal scaling - Tech stack behind reliable applications - Remote VCS repositories - Creating remote repositories - Git CLI	Lab, Lecture, Exercise & Supplementary Material
4	Developer Operations Landscape	1-3, 6	- DevOps roles & responsibilities - Tools & techniques in the DevOps sphere - DevOps culture and lifecycle - Core DevOps principles - Career opportunities in DevOps - Demand & compensation for engineers - VCS workflows - Working with Travis for CI - Configuring Travis - Generating test scripts - GitHub & Travis communications	Lab, Lecture, & Supplementary Material
5	Hosting High Geared Single Page Applications & Work tracking tools	1-6	- The anatomy of a Single Page App - Lightning fast deploys with SPAs - Hosting SPAs - Working with front-end developers - Front end tooling found for build sequencing - Introduction to build tools - Gradle and Maven - Work tracking using Jira	Lab, Lecture & Supplementary Material
6	Continuous Integration / Continuous Deployment	1-6	- Mechanisms to integrate and validate source code changes - Installing and configuring Jenkins for CI/CD pipeline - Deploying to infrastructure environments - How CI improves collaboration and code quality - Continuous Testing - How CI/CD increases deployment velocity	Lab, Lecture, Exercise & Supplementary Material

			- Working with markdown for GitHub documentation	
7	MID-TERM EXAM Mixed format test week 1-6			
8	INTERSESSION WEEK			
9	CI/CD Pipeline Sequences	1-6	- Building pipelines for failures - Monitoring pipeline sequences - Phases in a CD pipeline - Branch deployments in CI/CD pipelines - Running feature branches - Git hooks to notify changes to external apps - CI/CD tooling - Writing Declarative and Scripted pipeline in Jenkins - Achieving low risk releases with deployment strategies	Lab, Lecture, Exercise & Supplementary Material
10	Working with Cloud Infrastructure	1-7	- How does 'The Cloud' work? - Different types of cloud computing - Why the cloud matters for DevOps - XaaS and services supporting infrastructure - Popular examples of cloud computing - Cloud storage and how it's utilized through DevOps - Associated risks of cloud computing - Characteristics of Cloud computing - Analyzing the benefits of cloud computing - Serverless and cloud formation - Consider security working with cloud services / DevOps security best practices (SAST/DAST)	Lab, Lecture, Exercise & Supplementary Material
11	Automating DevOps & Quality assurance	1-7	- How much automation is too much? - Balancing tasks intelligently - Breaking down automation processes - Common use cases and problems solved - The role of performance testing - Test automation to enhance strategies - Introduction to BDD & TDD - Source Code Quality Check using SonarQube - Testing tools like Selenium and Appium - Test reports	Lab, Lecture, Exercise & Supplementary Material -
12	DevOps Assembly Lines	1-7	- Differentiating between CI pipelines and assembly lines - Managing and maintaining an assembly line - Maturing CI pipelines	Lab, Lecture, Exercise & Supplementary Material

			- Gluing activities together - Getting ready for AI and ML - DevOps & beyond	
13	Commonly Used DevOps Tech Exposed	1-7	- Remote server automation with Capistrano - Infrastructure Configuration, provisioning and Monitoring tools like Ansible, Puppet, Chef and Nagios - Visibility with New Relic - Working with ELK - Application containerization - Architecture and Components of Docker architecture - Writing Vagrant, YAML and Dockerfile scripts	Lab, Lecture, Exercise & Supplementary Material
14	Transitioning into the role of DevOps	1-8	- Getting a foot in the DevOps door - Focusing on the mindset of an engineer - Market demand for a DevOps engineer - Must have skills to penetrate DevOps market - Global market predictions - Average salary expectations for entry level positions - Typical interview questions - Additional resources & certification opportunities	Lab, Lecture & Supplementary material
15	Final exam			
For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx Policy on Academic Dishonesty: The <i>minimal</i> consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material. To view George Brown College policies please go to www.georgebrown.ca/policies				